■■ Bernoulli's Principle — The Flying Ball

Grade 1 Science Lesson Plan

Grade: 1	**Duration:** 45 Minutes	**Subject:** Science / Physics	**Language:** English (with Hindi cues)

■ Learning Objectives

- ✔ Understand that **air is real** and exerts force on objects.
- ✔ Recognise that air can move at **different speeds**.
- ✔ State Bernoulli's Principle in simple terms: **fast air pushes less** than slow air.
- ✔ Explain why a ball floats in a hair-dryer stream using Bernoulli's idea.
- ✔ Connect the principle to everyday Indian examples (cricket swing, kites, ceiling fans, aeroplanes).

■ Materials Needed

Demonstration	Student Activity (Try at Home)
Hair dryer (or a strong straw / pump)	Newspaper / scrap paper
Light ping-pong ball or paper ball	Drinking straw
Optional: cricket ball for discussion	Small bowl / katori
Whiteboard & markers	—

■ Lesson Structure (45 min)

Phase 1 — Hook & WOW! (5 min)

- Show the floating ball demo *without explanation*. Ask: *"Yeh magic hai? Yeh science hai!"*
- Invite predictions: *Why does the ball not fall down?*
- Build curiosity — tell students they will discover Bernoulli's secret today.

Phase 2 — Air is Real (8 min)

- Ask: *Can you see air? Can you feel it?* (Fan, blowing on hand)
- Discuss everyday examples: cooling hot chai, clothes drying on the terrace, kites on Makar Sankranti.
- Key message: **Air pushes things when it moves.**

Phase 3 — Fast Air vs. Slow Air (7 min)

- Demonstrate slow blow vs. strong blow on a feather.
- Introduce the **Busy Street Analogy**: Fast air = cars on a highway (less pushing). Slow air = traffic jam (more pushing).
- Have students say together: *"BER-NOO-LEE's Principle!"*
- Write on board: **Fast Air** → **Less Push | Slow Air** → **More Push**

Phase 4 — The Flying Ball Explained (10 min)

- Repeat demo; now explain step by step:
- ① The hair dryer blows **fast air** upward around the ball.
- ② Fast air pushes **less** than the slow air on the sides.
- ③ Slow air outside pushes the ball **back in** — so it floats!
- Draw a simple diagram on the board (ball, fast air arrows up, slow air arrows from sides).

Phase 5 — Bernoulli is Everywhere! (8 min)

- Cricket swing bowling: air moves differently on each side → ball curves.
- Kites on Makar Sankranti: wind lifts the patang up.
- Aeroplane wings: fast air above → less push → plane lifts off.
- Ceiling fan blades: spinning blades push cool air downward.
- Encourage students to spot more examples at home.

Phase 6 — Try It Yourself (5 min)

- Guide the home activity: crumple a newspaper ball, place in a katori, blow through a straw across the top.
- Students predict what will happen before trying.
- Discuss results: *Chalo! Did the ball move? Why?*

Phase 7 — Recap, Quiz & Badge (2 min)

- Quick recap — students answer together:
- **Q1:** Which air pushes more — slow or fast? *(Slow!)*
- **Q2:** Why does the ball float? *(Fast air pushes less; slow air outside pushes it back.)*
- Award everyone the ■ **Bernoulli Explorer** badge — *"Well done! Jai Hind!"*

■ Key Concept Summary

Bernoulli's Principle (BER-NOO-LEE)

Fast-moving air exerts less pressure than slow-moving air.
When fast air flows around a ball, the slower surrounding air pushes the ball back in — making it *float*!

■ Teacher Tips & Differentiation

■ **Safety:** Keep hair dryer on low heat / cool-air setting. Supervise carefully near young children.

■ **Scaffolding:** Use the traffic analogy repeatedly — it resonates with Indian urban children.

■ **Extension (advanced):** Ask: *How do aeroplane wings use this principle to fly?*

■ **ELL / multilingual support:** Use Hindi cues naturally (*yeh magic nahi, yeh science hai!*, *chalo!*) to build comfort.

■ **Cross-curricular link:** Connects to Maths (fast/slow, more/less) and Physical Education (cricket, kite flying).

■ Assessment

■ Formative (Verbal)	✏■ Summative (Written / Drawing)
Show floating ball demo again — can student explain what is happening?	Draw a ball floating in fast air — label fast air and slow air arrows.
Ask student to name one real-world example of Bernoulli's Principle.	Complete the sentence: *Fast air pushes* ________ *than slow air.*
Observation: does the student correctly distinguish fast vs. slow air pressure?	Circle the correct picture: which shows Bernoulli's Principle? (kite flying / rock on ground)

■ Vocabulary Wall

Air	The invisible gas all around us that we breathe and feel.
Pressure	The pushing force air (or anything) makes on a surface.
Bernoulli's Principle	Fast-moving air pushes less than slow-moving air.
Float	To stay up in the air or on water without sinking.
Speed	How fast something moves.
Swing (bowling)	When a cricket ball curves in the air due to air pressure differences.

■ Home Experiment — Try It Yourself!

Step 1 — Make a Ball	Crumple a piece of newspaper into a small, tight ball.
Step 2 — Gather Materials	Get a drinking straw and a small bowl (katori).
Step 3 — Set It Up	Place the newspaper ball inside the bowl.
Step 4 — Blow & Watch!	Hold the straw above the bowl edge and blow hard across the top. Watch the ball move! Chalo!
Step 5 — Explain It	Can you tell your family why the ball moved? Use Bernoulli's Principle!